Cal 2 - Exam Question 34

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Question 34

Evaluate

$$\int \frac{x^2}{(4-x^2)^{\frac{3}{2}}}\,dx.$$

a. $\dfrac{2x}{\sqrt{4-x^2}} - 2\sin^{-1}\left(\dfrac{x}{2}\right) + C$

b. $\dfrac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\dfrac{x}{2}\right) + C$

c. $\dfrac{4}{\sqrt{4-x^2}} - 2\sin^{-1}\left(\dfrac{x}{2}\right) + C$

d. I do not know

Solution: b) $\dfrac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\dfrac{x}{2}\right) + C$

This question is asking you to: Compute an antiderivative by rewriting the denominator into a form that matches a standard trigonometric substitution pattern, then substitute back to write the final answer in terms of x.

Step 1: Rewrite the denominator $(4-x^2)^{\frac{3}{2}}$.

Step 1.1:

Identify the denominator

In

$$\frac{x^2}{(4-x^2)^{\frac{3}{2}}},$$

the denominator is the bottom part:

$$(4-x^2)^{\frac{3}{2}}.$$

Step 1.2:

Rewrite fractional exponents

Since the denominator is complicated, we want to simplify it by rewriting the power.

$$(4-x^2)^{\frac{3}{2}}$$

has a fractional exponent, and we can rewrite fractional exponents using:

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m, \qquad \sqrt{a} = a^{\frac{1}{2}}.$$

Step 1.3:

Rewrite the power $\frac{3}{2}$

Rewrite $\frac{3}{2} = \frac{1}{2} \cdot 3$ $\quad (4-x^2)^{\frac{3}{2}} = (4-x^2)^{\frac{1}{2}\cdot 3}$

Use $a^{bc} = (a^b)^c$ $\quad (4-x^2)^{\frac{1}{2}\cdot 3} = \left((4-x^2)^{\frac{1}{2}}\right)^3$

Rewrite $(4-x^2)^{\frac{1}{2}}$ as a square root $\quad \left((4-x^2)^{\frac{1}{2}}\right)^3 = \left(\sqrt{4-x^2}\right)^3$

Step 2: Set up trigonometric substitution.

Step 2.1:

Notice the trig substitution pattern

Right now the denominator still looks complicated, so we want to simplify it more. The denominator contains

$$\sqrt{4-x^2},$$

and this looks like the trigonometric substitution pattern

$$\sqrt{a^2-x^2}.$$

Step 2.2:

Define the symbols

a: constant, $\quad x$: variable, $\quad \theta$: angle.

Step 2.3:

Trigonometric pattern $\sqrt{a^2 - x^2}$

When we encounter the pattern

$$\sqrt{a^2 - x^2},$$

we let

$$x = a \sin\theta.$$

Step 2.4:

Rewrite the fraction $\sqrt{a^2 - x^2}$

Substitute $x = a\sin\theta$ $\quad \sqrt{a^2 - x^2} = \sqrt{a^2 - (a\sin\theta)^2}$

Step 2.5:

Simplify $\sqrt{a^2 - (a\sin\theta)^2}$

Use Power of a Product Rule, $\quad (a\,b)^2 = a^2\,b^2$

Substitute $b = \sin\theta$, $\quad (a\,\sin\theta)^2 = a^2(\sin\theta)^2$

$$\sqrt{a^2 - (a\sin\theta)^2} = \sqrt{a^2 - a^2(\sin\theta)^2}$$

Step 2.6:

Trig substitution summary

For pattern $\sqrt{a^2 - x^2}$, let $x = a\sin\theta$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2(\sin\theta)^2}$$

Step 3: Rewrite $\sqrt{4 - x^2}$ using trigonometric substitution.

Step 3.1:

Identify a and x in $\sqrt{4 - x^2}$

Rewrite $4 = 2 \cdot 2 = 2^2$ $\quad \sqrt{4 - x^2} = \sqrt{2^2 - x^2}$

Let $\quad \sqrt{2^2 - x^2} = \sqrt{a^2 - x^2}$

$$a = 2$$

$$x = a\sin\theta = 2\sin\theta$$

Step 3.2:

Substitute $a = 2$, $x = 2\sin\theta$ **into the fraction**

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2(\sin\theta)^2}$$
$$\sqrt{2^2 - x^2} = \sqrt{2^2 - 2^2(\sin\theta)^2}$$

Step 4: Simplify $\sqrt{2^2 - 2^2(\sin\theta)^2}$**.**

Step 4.1:

Replace $2^2 = 4$

$$\text{Replace } 2^2 = 4 \quad \sqrt{2^2 - 2^2(\sin\theta)^2} = \sqrt{4 - 4(\sin\theta)^2}$$

Step 4.2:

Factor out 4

$$\text{Rewrite } 4 = 4 \cdot 1 \quad 4 - 4(\sin\theta)^2 = 4(1) - 4((\sin\theta)^2)$$
$$\text{Factor out } 4 \quad 4 - 4(\sin\theta)^2 = 4(1 - (\sin\theta)^2)$$

Step 4.3:

Simplify $1 - (\sin\theta)^2$

$$\text{By } (\sin\theta)^2 = \sin^2\theta, \quad 4(1 - (\sin\theta)^2) = 4(1 - \sin^2\theta)$$
$$\text{By Pythagorean identity}, \quad 1 - \sin^2\theta = \cos^2\theta$$
$$\text{Substitute the Pythagorean identity}, \quad 4(1 - \sin^2\theta) = 4(\cos^2\theta)$$

Step 4.4:

Summary

$$\begin{aligned}
\sqrt{4 - x^2} &= \sqrt{2^2 - x^2} \quad \text{from Step 3.1} \\
&= \sqrt{2^2 - 2^2(\sin\theta)^2} \quad \text{from Step 3.2} \\
&= \sqrt{4(1 - \sin^2\theta)} \quad \text{from Step 4.1 and Step 4.2} \\
&= \sqrt{4\cos^2\theta} \quad \text{from Step 4.3}
\end{aligned}$$

Step 5: Compute $(4 - x^2)^{\frac{3}{2}}$ **after substitution.**

Step 5.1:

Substitute into $\left(\sqrt{4-x^2}\right)^3$

$$\text{From Step 1.3,} \quad (4-x^2)^{\frac{3}{2}} = \left(\sqrt{4-x^2}\right)^3$$

$$\text{Substitute } \sqrt{4-x^2} = \sqrt{4\cos^2\theta} \quad \left(\sqrt{4-x^2}\right)^3 = \left(\sqrt{4\cos^2\theta}\right)^3$$

Step 5.2:

Simplify $\sqrt{4\cos^2\theta}$

$$\text{Since } \sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{4\ \cos^2\theta} = \sqrt{4}\,\sqrt{\cos^2\theta}$$

$$\sqrt{4} = 2 \quad \sqrt{4}\,\sqrt{\cos^2\theta} = 2\,\sqrt{\cos^2\theta}$$

Step 5.3:

Rewrite $\sqrt{4\cos^2\theta}$ as $2|\cos\theta|$

$$\text{Use } \sqrt{u^2} = |u| \quad \sqrt{\cos^2\theta} = |\cos\theta|$$

$$\text{Multiply both sides by 2} \quad 2\,\sqrt{\cos^2\theta} = 2\,|\cos\theta|$$

Step 5.4:

Cube both sides

$$\text{Cube both sides} \quad \left(\sqrt{4\cos^2\theta}\right)^3 = (2|\cos\theta|)^3$$

$$\text{Use } (ab)^3 = a^3b^3 \quad (2|\cos\theta|)^3 = 2^3\,|\cos\theta|^3$$

$$\text{Simplify } 2^3 \quad \left(\sqrt{4\cos^2\theta}\right)^3 = 8\,|\cos\theta|^3$$

Step 5.5:

Remove the absolute value

$$\text{For } x = 2\sin\theta, \text{ we take } 0 \le \theta \le \tfrac{\pi}{2} \quad \cos\theta \ge 0$$

$$\text{So } |\cos\theta| = \cos\theta \quad 8|\cos\theta|^3 = 8\cos^3\theta$$

Step 5.6:

Summary

$$\sqrt{4\cos^2\theta} = 2\sqrt{\cos^2\theta} \quad \text{from Step 5.2}$$
$$2\sqrt{\cos^2\theta} = 2|\cos\theta| \quad \text{from Step 5.3}$$
$$\left(\sqrt{4\cos^2\theta}\right)^3 = 8|\cos\theta|^3 \quad \text{from Step 5.4}$$
$$(4-x^2)^{\frac{3}{2}} = 8\cos^3\theta \quad \text{from Step 5.6}$$

Step 6: Rewrite the numerator x^2.

Step 6.1:

Square both sides

$$\text{Start with the substitution} \quad x = 2\sin\theta \quad \text{from Step 3.1}$$
$$\text{Square both sides} \quad x^2 = (2\sin\theta)^2$$

Step 6.2:

Apply $(ab)^2 = a^2b^2$

$$\text{Let } a = 2,\ b = \sin\theta \quad (ab)^2 = a^2b^2$$
$$\text{Substitute into the rule} \quad (2\sin\theta)^2 = 2^2(\sin\theta)^2$$

Step 6.3:

Simplify $2^2(\sin\theta)^2$

$$\text{Compute } 2^2 \quad 2^2 = 4$$
$$\text{By square of trigonometric functions} \quad (\sin\theta)^2 = \sin^2\theta$$
$$\text{Substitute} \quad 2^2(\sin\theta)^2 = 4\sin^2\theta$$
$$\text{Therefore} \quad x^2 = 4\sin^2\theta$$

Step 7: Compute dx.

Step 7.1:

Differentiate $x = 2\sin\theta$

$$\begin{aligned}
\text{Start with the substitution} \quad & x = 2\sin\theta \quad \text{from Step 3.1} \\
\text{Differentiate both sides with respect to } \theta \quad & \tfrac{dx}{d\theta} = \tfrac{d}{d\theta}(2\sin\theta) \\
\text{Use the constant multiple rule} \quad & \tfrac{dx}{d\theta} = 2\,\tfrac{d}{d\theta}(\sin\theta) \\
\text{Since } \tfrac{d}{d\theta}(\sin\theta) = \cos\theta \quad & \tfrac{dx}{d\theta} = 2\cos\theta
\end{aligned}$$

Step 7.2:

Write dx

$$\begin{aligned}
\text{Start with} \quad & \tfrac{dx}{d\theta} = 2\cos\theta \\
\text{Multiply both sides by } d\theta \quad & d\theta\left(\tfrac{dx}{d\theta}\right) = (2\cos\theta)\,d\theta \\
\text{So} \quad & dx = 2\cos\theta\,d\theta
\end{aligned}$$

Step 8: Rewrite the integral with substitutions.

Step 8.1:

List substitutions

$$\begin{aligned}
x^2 &= 4\sin^2\theta \quad \text{from Step 6} \\
(4-x^2)^{\frac{3}{2}} &= 8\cos^3\theta \quad \text{from Step 1 to Step 5} \\
dx &= 2\cos\theta\,d\theta \quad \text{from Step 7}
\end{aligned}$$

Step 8.2:

Substitute

$$\begin{aligned}
\text{Start with the original integral} \quad & \int \frac{x^2}{(4-x^2)^{\frac{3}{2}}}\,dx \\
\text{Substitute the expressions from Step 8.1} \quad & \int \frac{x^2}{(4-x^2)^{\frac{3}{2}}}\,dx = \int \frac{4\sin^2\theta}{8\cos^3\theta}\,(2\cos\theta\,d\theta)
\end{aligned}$$

Step 9: Simplify the integrand.

Step 9.1:

Multiply constants

$$\int \frac{4\sin^2\theta}{8\cos^3\theta}\, 2\cos\theta \, d\theta = \int \frac{4\cdot 2\,\sin^2\theta\,\cos\theta}{8\cos^3\theta}\, d\theta$$

Simplify $4\cdot 2$ $$\int \frac{4\cdot 2\sin^2\theta\cos\theta}{8\cos^3\theta}\, d\theta = \int \frac{8\,\sin^2\theta\cos\theta}{8\cos^3\theta}\, d\theta$$

Step 9.2:

Cancel 8

Cancel the common factor 8 $$\int \frac{8\,\sin^2\theta\cos\theta}{8\,\cos^3\theta}\, d\theta = \int \frac{\sin^2\theta\cos\theta}{\cos^3\theta}\, d\theta$$

Step 9.3:

Cancel $\cos\theta$

Rewrite $\cos^3\theta$ $$\cos^3\theta = \cos^2\theta\cos\theta$$

Substitute into the denominator $$\int \frac{\sin^2\theta\cos\theta}{\cos^3\theta}\, d\theta = \int \frac{\sin^2\theta\cos\theta}{\cos^2\theta\cos\theta}\, d\theta$$

Cancel $\cos\theta$ $$\int \frac{\sin^2\theta\,\cos\theta}{\cos^2\theta\,\cos\theta}\, d\theta = \int \frac{\sin^2\theta}{\cos^2\theta}\, d\theta$$

Step 9.4:

Simplify $\int \frac{\sin^2\theta}{\cos^2\theta}\, d\theta$

By definition of tangent, $$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Square both sides $$\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$$

Rewrite the integrand $$\int \frac{\sin^2\theta}{\cos^2\theta}\, d\theta = \int \tan^2\theta\, d\theta$$

Step 10: Integrate in θ.

Step 10.1:

Rewrite the integral

Use the trigonometry identity	$\tan^2\theta = \sec^2\theta - 1$
Substitute into the integral	$\int \tan^2\theta \, d\theta = \int (\sec^2\theta - 1) \, d\theta$
By additive property of integrals	$\int (\sec^2\theta - 1) \, d\theta = \int \sec^2\theta \, d\theta - \int 1 \, d\theta$

Step 10.2:

Integrate

Integrate $\sec^2\theta$	$\int \sec^2\theta \, d\theta = \tan\theta$
Integrate 1	$\int 1 \, d\theta = \theta$
Combine the result	$\int \sec^2\theta \, d\theta - \int 1 \, d\theta = \tan\theta - \theta + C$
Therefore	$\int \tan^2\theta \, d\theta = \tan\theta - \theta + C$

Step 11: Convert $\tan\theta - \theta + C$ back into x.

Step 11.1:

Identify what to convert back We now have the antiderivative in terms of θ, and we want to convert it back to x. To do that, we need to rewrite both $\tan\theta$ and θ in terms of x.

Step 11.2:

Find θ

Start with	$x = 2\sin\theta$ from Step 3.1
Multiply both sides by $\frac{1}{2}$	$\frac{1}{2}x = \frac{1}{2}2\sin\theta$
$\frac{1}{2}\cdot 2 = 1$	$\frac{x}{2} = \sin\theta$
Apply $\sin^{-1}$ to both side of the equation	$\theta = \sin^{-1}\left(\frac{x}{2}\right)$

Step 11.3:

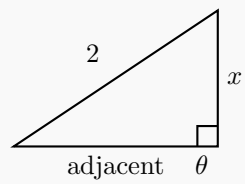
Write the sine ratio

Use $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$ $\quad \sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{x}{2}$

Step 11.4:

Draw the triangle and identify the sides

From $\sin\theta = \frac{x}{2}$ opposite $= x$, hypotenuse $= 2$



Step 11.5:

Find the adjacent side

By the Pythagorean Theorem, adjacent $= \sqrt{\text{hypotenuse}^2 - \text{opposite}^2}$

Substitute the values adjacent $= \sqrt{2^2 - x^2} = \sqrt{4 - x^2}$

Step 11.6:

Find $\tan\theta$

Use $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$ $\tan\theta = \dfrac{x}{\sqrt{4 - x^2}}$

Step 12: Write the final answer and choose the correct option.

Step 12.1:

Substitute back

From Step 10.2, $\displaystyle\int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}}\,dx = \tan\theta - \theta + C$

Substitute Step 11.2 and Step 11.6 $\displaystyle\int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}}\,dx = \frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$

Step 12.2:

Final answer

The correct option is b.

Definitions and Notation

Antiderivative

Definition: An antiderivative of a function is another function whose derivative gives back the original function. In simple words: if you take the derivative of the new function, you return to the old one.

Example:

$$\frac{d}{dx}\left(\frac{x^3}{3}\right) = x^2,$$

so one antiderivative of x^2 is $\frac{x^3}{3}$.

Denominator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the denominator is the bottom part, $q(x)$.

Example:

$$\frac{x^2}{(4-x^2)^{3/2}}$$

has numerator x^2 and denominator $(4-x^2)^{3/2}$.

Numerator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the numerator is the top part, $p(x)$.

Example:

$$\frac{x^2}{(4-x^2)^{3/2}}$$

has numerator x^2.

Constant

Definition: A constant is a fixed, unchanging value or number, such as 5, -10, or π, that remains the same throughout a specific expression, equation, or function.

Example: In

$$3x + 7,$$

the 7 is the constant term.

Variable

Definition: A variable is a symbol, typically written in a lowercase letter like x, y, or z, that represents an unknown numerical value or a quantity that can change.

Example: In

$$3x + 7,$$

the x is the variable.

Power of a Product Rule

Definition: When raising a product of terms to a power, apply that power to each factor inside the parentheses, expressed as

$$(ab)^n = a^n b^n.$$

Example:

$$(3x)^2 = 3^2 x^2 = 9x^2.$$

Trigonometric Substitution

Definition: Trigonometric substitution is a method for simplifying integrals with square roots by replacing x with a trig expression involving an angle.

Example:

$$\sqrt{4 - x^2} = \sqrt{2^2 - x^2}$$

matches the pattern $\sqrt{a^2 - x^2}$, so we use $x = 2\sin\theta$.

Common patterns:

$$\sqrt{a^2 - x^2} : x = a\sin\theta, \qquad \sqrt{a^2 + x^2} : x = a\tan\theta, \qquad \sqrt{x^2 - a^2} : x = a\sec\theta.$$

Fractional Exponents

Definition: Fractional exponents rewrite powers as roots.

$$a^{m/n} = \sqrt[n]{a^m} = \left(a^{1/n}\right)^m.$$

Example:

$$(4 - x^2)^{3/2} = \left((4 - x^2)^{1/2}\right)^3 = \left(\sqrt{4 - x^2}\right)^3.$$

This means the 3/2 power tells us to take a square root first, then cube.

Square of Trigonometric Functions

Definition:

$$(\sin\theta)^2 = \sin^2\theta, \qquad (\cos\theta)^2 = \cos^2\theta, \qquad (\tan\theta)^2 = \tan^2\theta.$$

Pythagorean Identity

Definition: The Pythagorean identity is the trig fact

$$\sin^2\theta + \cos^2\theta = 1 \quad \Longleftrightarrow \quad 1 - \sin^2\theta = \cos^2\theta.$$

Example:

$$4 - 4\sin^2\theta = 4(1 - \sin^2\theta) = 4\cos^2\theta.$$

Pythagorean Theorem

Definition: In a right triangle with legs a and b and hypotenuse c,

$$a^2 + b^2 = c^2.$$

Example: If the hypotenuse is 2 and one side is x, then the other side satisfies

$$\text{adjacent}^2 = 2^2 - x^2 = 4 - x^2.$$

Square Root

Definition:

$$y^2 = x \quad \Longrightarrow \quad y \text{ is a square root of } x.$$

For nonnegative real numbers,

$$\sqrt{x} = y \quad \Longleftrightarrow \quad y^2 = x \text{ and } y \geq 0.$$

Example:

$$\sqrt{9} = 3$$

because

$$3^2 = 9.$$

Key Properties:

$$\sqrt{x^2} = |x|$$

$$\sqrt{a}\sqrt{b} = \sqrt{ab}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\left(\sqrt{a}\right)^2 = a \quad \text{and} \quad a^{\frac{1}{2}} = \sqrt{a}$$

Constant Multiple Rule

Definition: When taking a derivative, a constant factor can stay in front:

$$\frac{d}{d\theta}\big(c\, f(\theta)\big) = c\, f'(\theta).$$

Example:

$$\frac{d}{d\theta}(2\sin\theta) = 2\cos\theta.$$

Additive Property of Integrals

Definition: The integral of a sum or difference can be split into separate integrals:

$$\int (f(x) + g(x))\, dx = \int f(x)\, dx + \int g(x)\, dx$$

and

$$\int (f(x) - g(x))\, dx = \int f(x)\, dx - \int g(x)\, dx.$$

Example:

$$\int (\sec^2\theta - 1)\, d\theta = \int \sec^2\theta\, d\theta - \int 1\, d\theta.$$

Absolute Value

Definition: The absolute value of a number is its distance from 0, so it is never negative.

Example:

$$|3| = 3, \qquad |-3| = 3.$$

In Q34 we use

$$\sqrt{u^2} = |u|.$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we know $\cos\theta \geq 0$, so

$$|\cos\theta| = \cos\theta.$$

Basic Trig Ratios: Sine, Cosine, and Tangent

Definition: In a right triangle:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Example: If $\sin\theta = \frac{x}{2}$, then opposite $= x$ and hypotenuse $= 2$. The adjacent side is

$$\sqrt{2^2 - x^2} = \sqrt{4 - x^2},$$

so

$$\tan\theta = \frac{x}{\sqrt{4 - x^2}}.$$

Inverse Sine $\sin^{-1}(x)$

Definition: Also written $\arcsin(x)$, inverse sine returns the angle whose sine is x.

Example: If $\sin\theta = \frac{x}{2}$, then

$$\theta = \sin^{-1}\left(\frac{x}{2}\right).$$